ABCjs - Template

This document defines some basic macros for that allow you to write music notes with the ABC-notation. To transform this into nicely rendered music notes an to play the music online as well, this plugin uses the JavaScript library abcjs. This plugin allows you to use ABC music in two ways, only for showing and playing, or to write and interpret ABC interactively with an editor.

Try it on LiaScript:

https://liascript.github.io/course/?
https://raw.githubusercontent.com/liaTemplates/ABCjs/main/README.md

See the project on Github:

https://github.com/liaTemplates/ABCjs

There are three ways to use this template. The easiest way is to use the `import` statement and the url of the raw text-file of the master branch or any other branch or version. But you can also copy the required functionality directly into the header of your Markdown document, see therefor the last slide. And of course, you could also clone this project and change it, as you wish.

1. Load the macros via

   `import: https://raw.githubusercontent.com/liaTemplates/ABCjs/main/README.md`

   or use this specific version and you course will be stable:

   `import: https://raw.githubusercontent.com/LiaTemplates/ABCjs/0.0.2/README.md`

2. Copy the definitions into your Project
3. Clone this repository on GitHub

`@ABCJS.render`

Display the ABC defined music with default options, which will render them as musical notes and add a the possibility to play the music aloud.
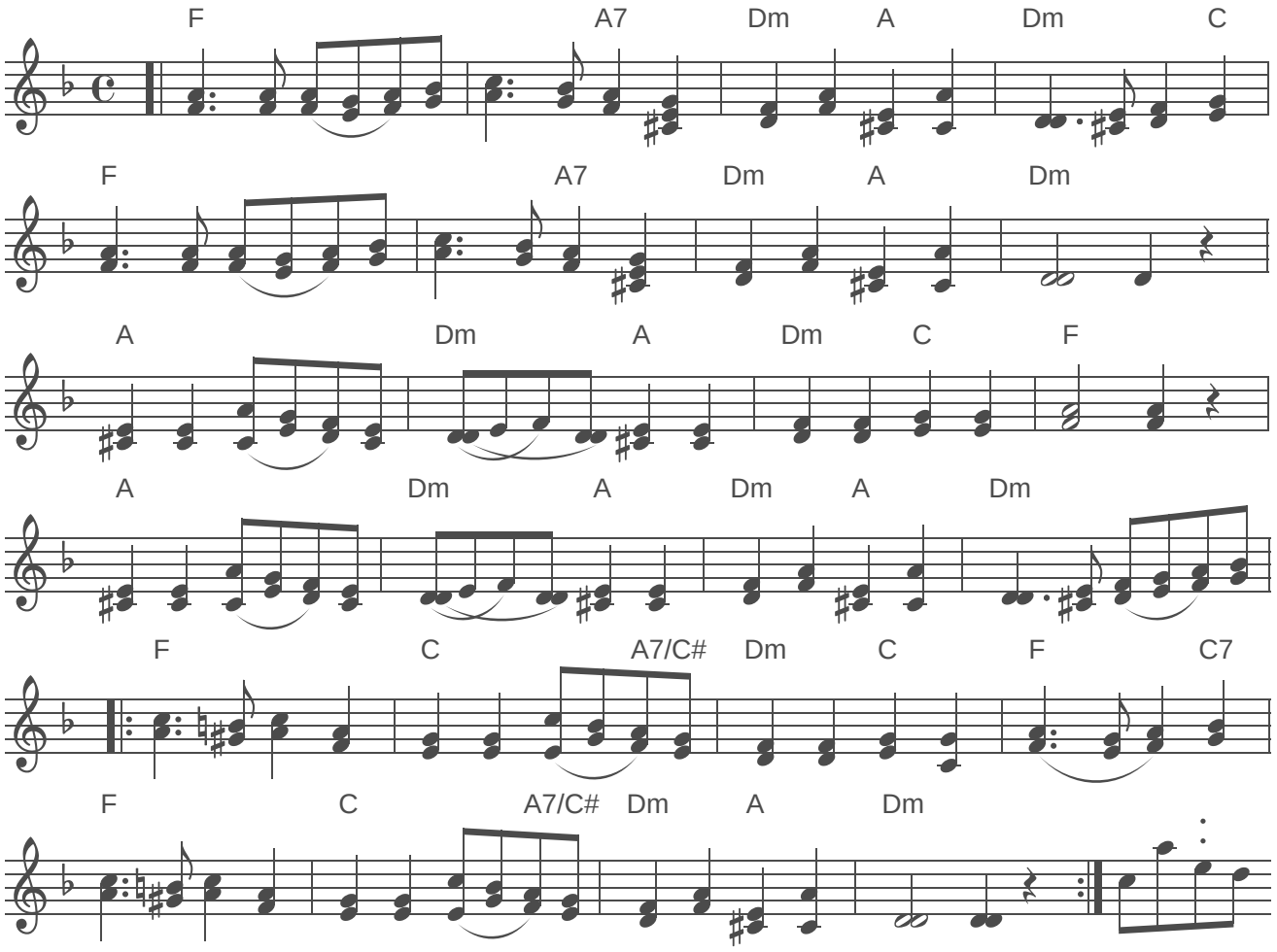
```
``` abc  @ABCJS.render
X: 1
T: Shche ne Vmerla Ukrayiny ni Slava, ni volya
T: Ukrainian National Anthem
C: trad.
R: march
S: https://www.8notes.com/scores/34640.asp
Z: 2022 John Chambers <jc:trillian.mit.edu>
M: C
L: 1/8
```
```

```
K: F
[|\
"F"[A3F3][AF] ([AF][GE][AF])[BG] | [c3A3][BG] "A7"[A2F2][G2E2^C2] |
"Dm"[F2D2][A2F2] "A"[E2^C2][A2C2] | "Dm"[D3D3][E^C] [F2D2]"C"[G2E2] |
"F"[A3F3][AF] ([AF][GE][AF])[BG] | [c3A3][BG] "A7"[A2F2][G2E2^C2] |
"Dm"[F2D2][A2F2] "A"[E2^C2][A2C2] | "Dm"[D4D4] D2z2 |
"A"[E2^C2][E2C2] ([AC][GE][FD])[EC] | ("Dm"[DD3-]EF)[DD] "A"[E2^C2][E2C2] |
"Dm"[F2D2][F2D2] "C"[G2E2][G2E2] | "F"[A4F4] [A2F2]z2 |
"A"[E2^C2][E2C2] ([AC][GE][FD])[EC] | ("Dm"[DD3-]EF)[DD] "A"[E2^C2][E2C2] |
"Dm"[F2D2][A2F2] "A"[E2^C2][A2C2] | "Dm"[D3D3][E^C] ([FD][GE][AF])[BG] |
|:\
"F"[c3A3][=B^G] [c2A2][A2F2] | "C"[G2E2][G2E2] ([cE][BG]"^A7/C#"[AF])[GE] |
"Dm"[F2D2][F2D2] "C"[G2E2][G2C2] | ("F"[A3F3][GE][A2F2])"C7"[B2G2] |
"F"[c3A3][=B^G] [c2A2][A2F2] | "C"[G2E2][G2E2] ([cE][BG]"^A7/C#"[AF])[GE] |
"Dm"[F2D2][A2F2] "A"[E2^C2][A2C2] | "Dm"[D4D4] [D2D2]z2 :|
```

Shche ne Vmerla Ukrayiny ni Slava, ni volya

Ukrainian National Anthem

march *trad.*



Source: https://www.8notes.com/scores/34640.asp
Transcription: 2022 John Chambers <jc:trillian.mit.edu>

0:00 100 % (180 BPM)

@ABCJS.renderWith

Use the render with option to change the default settings. A list of all possible parameters is defined in section Options. Additionally it is possible to set these parameters as comments within the ABC-code, as it is demonstrated here.

```
``` abc  @ABCJS.renderWith(channel="10" audio="true" notes="false")
X: 1
T: Cooley's
M: 4/4
L: 1/8
K: Emin
```
```

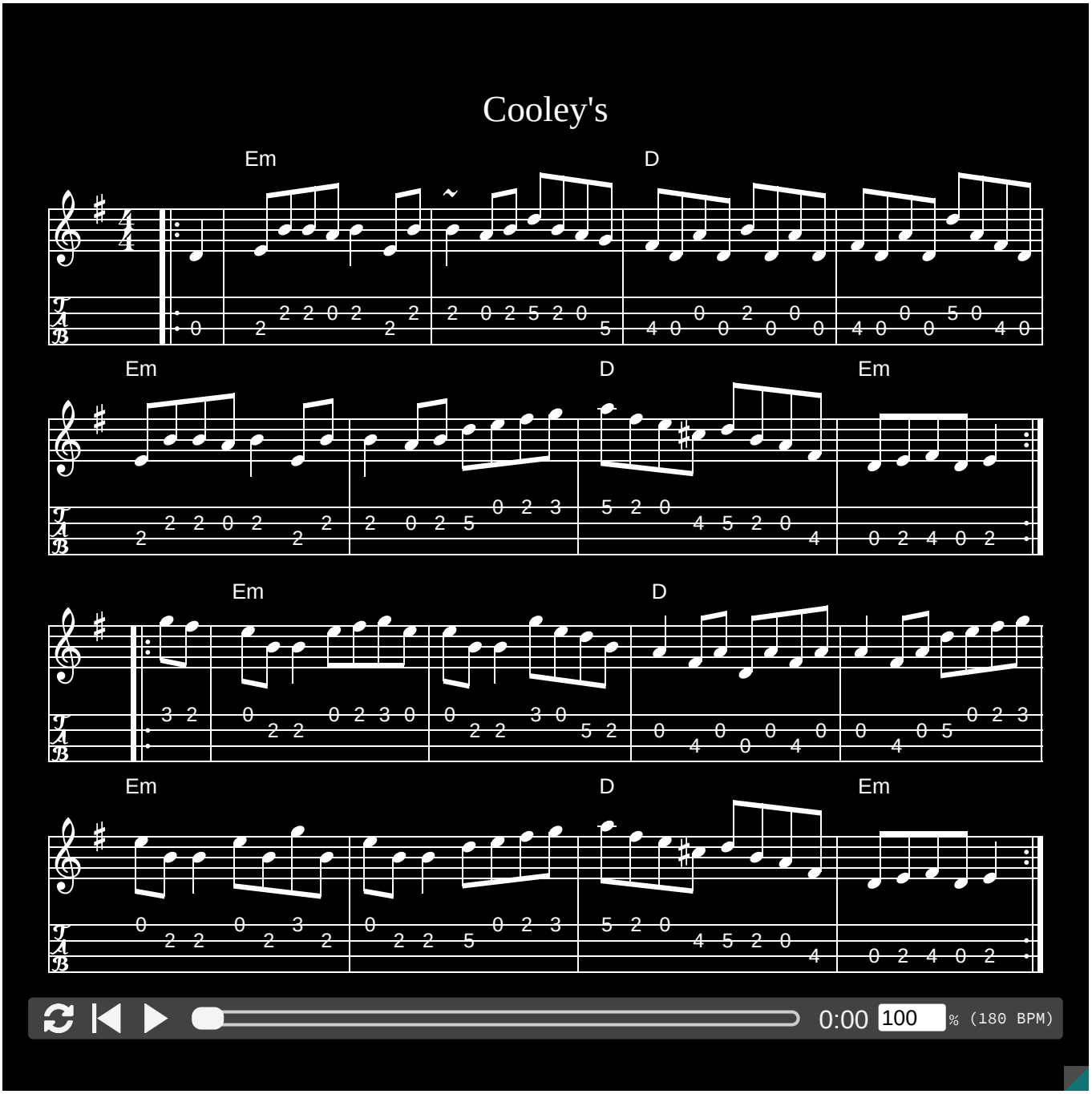
```
|:D2|"Em"EBBA B2 EB|~B2 AB dBAG|"D"FDAD BDAD|FDAD dAFD|
"Em"EBBA B2 EB|B2 AB defg|"D"afe^c dBAF|"Em"DEFD E2:|
|:gf|"Em"eB B2 efge|eB B2 gedB|"D"A2 FA DAFA|A2 FA defg|
"Em"eB B2 eBgB|eB B2 defg|"D"afe^c dBAF|"Em"DEFD E2:|
```



`@ABCJS.eval`

Next to just showing the notes, it is also possible to directly edit music with the LiaScript editor. Alls states and versions will be preserved. Simply add the `@ABCJS.eval` macro to the end of the code-block. This will run `eval` with the default settings and print out the result within the console. The code-example contains a list of all settings that can be changed by the user, this will overwrite the default settings and gives you more control over your composition.

```
% audio: true
% autoplay: false
% channel: 0
% debug: false
% notes: true
% program: 60
% responsive: true
% tablature: [{"instrument": "violin","tuning": ["G,", "D", "A", "e"]
% voicesOff: false
% chordsOff: false
% stereo: true
X: 1
T: Cooley's
M: 4/4
L: 1/8
K: Emin
|:D2|"Em"EBBA B2 EB|~B2 AB dBAG|"D"FDAD BDAD|FDAD dAFD|
"Em"EBBA B2 EB|B2 AB defg|"D"afe^c dBAF|"Em"DEFD E2:|
|:gf|"Em"eB B2 efge|eB B2 gedB|"D"A2 FA DAFA|A2 FA defg|
"Em"eB B2 eBgB|eB B2 defg|"D"afe^c dBAF|"Em"DEFD E2:|
```

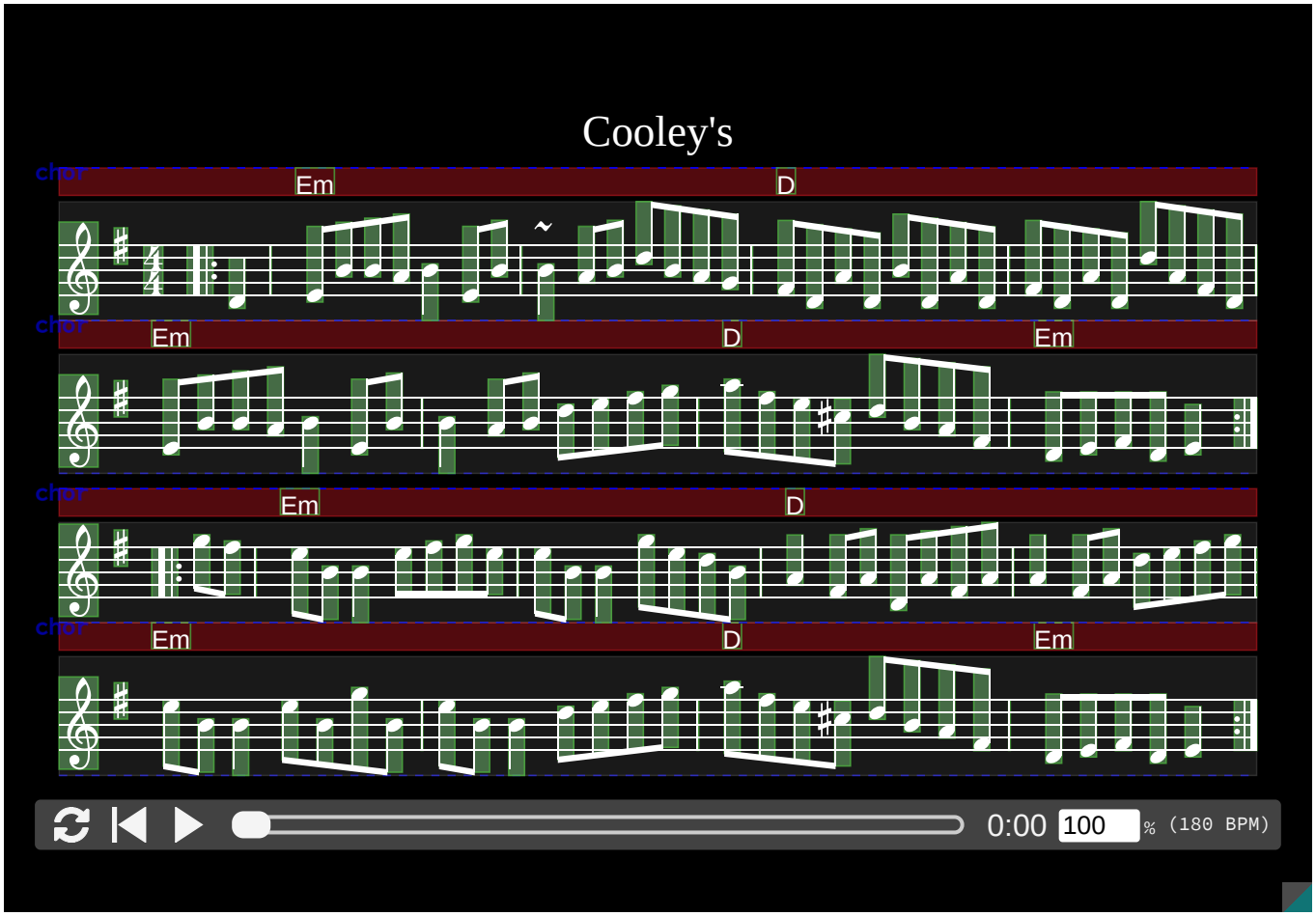


`@ABCJS.evalWith`

This is similar to the `@ABCJS.renderWith` version an allows you to configure the output more precisely. Additionally it is also possible, to overwrite these settings within the comments, as shown in the previous section.

```
X: 1
T: Cooley's
M: 4/4
L: 1/8
K: Emin
|:D2|"Em"EBBA B2 EB|~B2 AB dBAG|"D"FDAD BDAD|FDAD dAFD|
```

```
"Em"EBBA B2 EB|B2 AB defg|"D"afe^c dBAF|"Em"DEFD E2:|
|:gf|"Em"eB B2 efge|eB B2 gedB|"D"A2 FA DAFA|A2 FA defg|
"Em"eB B2 eBgB|eB B2 defg|"D"afe^c dBAF|"Em"DEFD E2:|
```



Options

option	Description
`audio`	Display audio settings, by default true.
`autoplay`	Start playing immediately when the site/abc becomes visible (by default disabled)
`channel`	The “midi channel” to use. This isn't particularly useful except that specifying channel 10 means to use the percussion sounds.
`debug`	Combinations of ABCJS `showDebug: ['box', 'grid']` which shows some visual debug information.
`notes`	Depict the notes, by default this is enabled, set it to false to hide them and to display only the “playback”.
`program`	A number for the midi program (aka “instrument”) to use, if not specified in ABC string.
`responsive`	Used by default, if you set this to false, then there will be a scrollbar displayed on small screens.
`tablature`	Add tablature options as defined here.
`voicesOff`	If true, play the metronome and accompaniment; do the animation callbacks, but don't play any melody lines.
`chordsOff`	If true, then don't turn the guitar chord symbols into sound. (But do play the metronome if there is one.)
`stereo`	Use pan to mimic stereo sound (by default set to false).

ABC Tutorial

You can search https://abcnotation.com for a vast amount of songs and resources. Additionally, it might be helpful to have a look at the following videos:

ABC - How to get started with abc notation

Chris Walshaw





Watch on

ABC - How to understand abc notation ... the basics

Chris Walshaw





Watch on

ABC - How to understand abc notation ... next steps

Chris Walshaw





Watch on

ABC - How to understand abc notation ... next steps

Chris Walshaw





Watch on

ABC - How to include lyrics in abc notation
Chris Walshaw



Watch on

ABC - How to write abc notation for multiple voices (polyphony)
Chris Walshaw



Watch on

Implementation

```
script:   https://cdn.jsdelivr.net/gh/liatemplates/abcjs@0.0.2/dist/index.j

@ABCJS.render: @ABCJS.renderWith( ,```@0```)

@ABCJS.renderWith
```

```
<lia-keep>
<lia-abcjs @0>@1</lia-abcjs>
</lia-keep>
@end

@ABCJS.eval: @ABCJS.evalWith( ,@0)

@ABCJS.evalWith
<script>
console.html(`<lia-abcjs @0>@input</lia-abcjs>`)
"LIA: stop"
</script>
@end
```